

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2013-14

Ancillary Botany-I**Paper: BOB-203A**

Time: Three Hours

Maximum Marks: 70

Note: Attempt five questions in all, including question no. 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Fill in the blanks or answer the following in not more than 50 words each: [10]

- (a) Why is RNA considered as the progenitor for the origin of first cellular life on the earth?
- (b) How old is the Earth?
- (c) 'Three-domain' system of classification was proposed by _____.
- (d) Heterocyst is the site of _____.
- (e) Name two microbes used as food.
- (f) Mention a typical feature to differentiate the Angiosperms from the Gymnosperms.
- (g) What is idioandrosporous?
- (h) What is the function of elater?
- (i) What is a secondary pollutant?
- (j) Define wet deposition.

Question 2. Write short notes on any two of the following: [15]

- (a) Microorganisms capable of producing antibiotics
- (b) Cyanobacteria as biofertilizer
- (c) Paper / Bacterial endospores

Question 3. Give a brief account of the classification of microbes as proposed by Whittaker. [15]

Question 4. Briefly describe the structure and asexual reproduction of *Oedogonium*. [15]

Question 5. Draw the labelled diagram of any two of the following (description not required): [15]

- (a) T.S. stem of *Selaginella*
- (b) *Marchantia* sporophyte

(c) Thallus structure of *Rhizopus*

Question 6. Briefly describe the reproduction in Gymnosperms with special reference to *Cycas*. [15]

Question 7. Give an account of air pollutants and their sources. Suggest suitable methods for controlling air pollution. [15]

Question 8. Define the sources of greenhouse gases in the atmosphere. Explain the greenhouse effect and its impact on climate change. [15]

Question 9. Write short notes on any two of the following: [15]

(a) V.S. archegoniophore of *Marchantia*

(b) Acid rain

(c) Threats to biodiversity

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